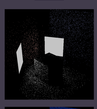
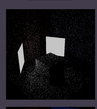
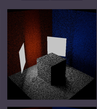
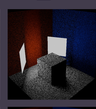
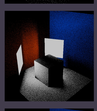
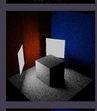
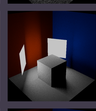
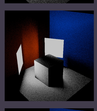
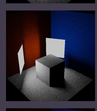
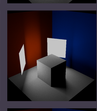
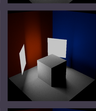
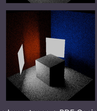
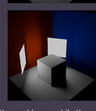
ALFie
Area Light Framework

Direct Lighting
***Lambert*

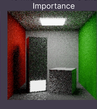
	Uniform	PDF Cosine	Area importance	MIS*
1 Ray				
50 Rays				
128 Rays				
256 Rays				

* MIS is combining Area Importance + PDF Cosine and Area Importance is the workhorse while the PDF Cosine contribution is marginal, hence the low gain compared with Area Importance sampling
** MIS Area Importance + PDF Cosine works well with indirect lighting as seen bellow in Lambert + GGX, 1 bounce Cornell Box

10 Cosine light bounces

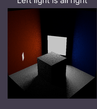
10 Area Importance no bounce

10 MIS from Cosine and Importance

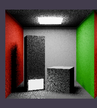
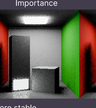


Area uniform
left small light is over powered

Area importance
w = ΣAi*Li
Left light is all right

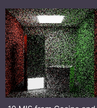
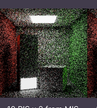
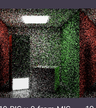
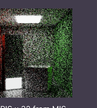


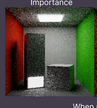
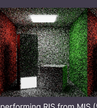
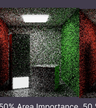
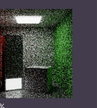
Resampled importance sampling
10 rays direct Lambert + GGX

10 Area importance	10 RIS x 2 from Importance	10 RIS x 8 from Importance	10 RIS x 32 from Importance
			

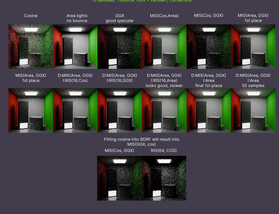
Specular more stable

10 rays 1 bounce, Lambert + GGX

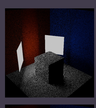
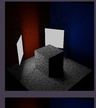
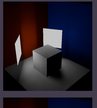
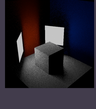
10 Cosine	10 RIS x 2 from Cos	10 RIS x 8 from Cos	10 RIS x 32 from Cos
			

10 MIS from Cosine and Imporance	10 RIS x 2 from MIS	10 RIS x 8 from MIS	10 RIS x 32 from MIS
			

When performing RIS from MIS (50% Area Importance, 50 % Cosine), mostly Cosine survives the collapsing



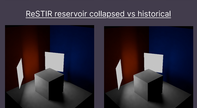
Resampled sampling
ReSTIR-lite, one ray per pixel, direct lighting, Lambert
(lite means no RIS, historical data is integrated)

	0×0 Spatial Kernel	7×7 Spatial Kernel	15×15 Spatial Kernel
Temporal budget 1 No history			
Temporal budget 2			
Temporal budget 4			
Temporal budget 8			

Spatial samples Russian Roulette

	0% samples dropped	25% samples dropped	50% samples dropped	90% samples dropped
Temporal budget 1 7 × 7 Spatial Kernel				
Temporal budget 4 7 × 7 Spatial Kernel				
Temporal budget 8 7 × 7 Spatial Kernel				

ReSTIR reservoir collapsed vs historical



ReSTIR spatial vs temporal

ReSTIR Historical

Maximum of 8 temporal samples are kept in the reservoir, allowing recomputation of PDF and cos θ.

ReSTIR pure

Collapsed reservoir. Only the reservoir champion sample is kept.

Spatial merging of 50 samples with visibility resampled

Temporal budget 1
7×7 Spatial Kernel

Temporal merging of 50 samples

Temporal budget 50
0×0 Spatial Kernel

No major differences noticed

Temporal resampling dominates

ReSTIR DI

ReSTIR 1 ray per pixel
BDRF: Lambert + GGX (different roughness)
Init samples 5 MIS(AreaLight, GGX)
Temporal accumulation: 4 frames
Spatial accumulation: 12 samples

ReSTIR Lambert spatial study

BDRF: Lambert
Init samples 5 x Area Light Importance
Temporal accumulation: 2 frames
Spatial accumulation: 5 samples radius 10

no restriction on join

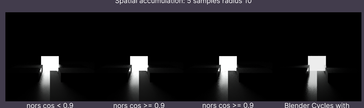
nors cos >= 0.9
fireflies on box edge removed

nors cos >= 0.9
pos dist < 0.04
better shadow
lower convergence

Blender Cycles with 10 rays per pixel, no denoise, no indirect

ReSTIR GGX spatial study

BDRF: GGX (different roughness)
Init samples 5 x MIS(AreaLight, GGX)
Temporal accumulation: 2 frames
Spatial accumulation: 5 samples radius 10



nors cos < 0.9
pos dist < 0.04
Notice glosy cube on the right of area light.
Speculars spreads from spatial join

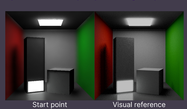
nors cos >= 0.9
pos dist < 0.04
halfs cos >= 0.9
Adding restriction between half vector norrows the specular

nors cos >= 0.9
pos dist < 0.04
halfs cos >= 0.98
better specular
lower convergence

Blender Cycles with 10 rays per pixel, no denoise, no indirect

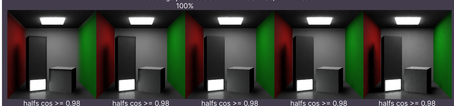
Average numbers of samples merged per pixel using 5 initial samples; temporal 2 frames, spatial 5 samples with radius 10:
73.51 [100%] - no restriction on spatial join
57.97 [78%] - pos dist < 0.1
35.07 [47%] - pos dist < 0.04 [good shadows]
33.12 [45%] - pos dist < 0.1 and nors cos >= 0.0 [good shadows & good lambert]
26.35 [35%] - pos dist < 0.1 and nors cos >= 0.9 and halfs cos >= 0.98 [good shadows, good lambert, good ggx]
49.41 [67%] - halfs cos >= 0.98

Lambert + GGX, 5 x MIS(Area, GGX), Temporal 4, Spatial 12 from radius 10
Visual impact & reservoir join percentage vs join restriction



Start point
no restriction on spatial
average join: 13583.87
100%

Visual reference
Cycles 10 samples
no denoise, no indirect



halfs cos >= 0.98
average join: 6768.61
49.82%

halfs cos >= 0.98
nors cos > 0.9
average join: 6478.03
47.68%

halfs cos >= 0.98
nors cos > 0.9
pos dist < 0.5
average join: 6469.80
47.62%

halfs cos >= 0.98
nors cos > 0.9
pos dist < 0.1
average join: 3692.08
27.17%

halfs cos >= 0.98
nors cos > 0.9
pos dist < 0.04
average join: 190.19
1.40%

- Artefacts: nors cos >= 0.9 and pos dist < 0.5 is needed to reduce artefacts

- Lambert: overall it looks good everywhere (once the artefacts are removed)

- GGX: halfs cos >= 0.98 results in good specular compared with Cycles and we drop half the spatial samples

- Shadows: best shadows compared with Cycles, results indeed with pos dist < 0.04 (small box shadow is not that rounded) which also yields the worse reservoir joining 1.4%. Acceptable shadows begin from pos dist < 0.1

Lambert + GGX, 5 x MIS(Area, GGX), Temporal 4, Spatial 12 samples - no halfs restriction
Visual impact & reservoir join percentage vs area concentration

Radius 10 - Area 314 reservoirs

Start point
no restriction on spatial
average join: 13583.87
100%

nors cos > 0.9
pos dist < 0.5
average join: 12783.13
47.62%

nors cos > 0.9
pos dist < 0.1
average join: 6833.63
2717%

nors cos > 0.9
pos dist < 0.04
average join: 334.35
1.40%

Radius 5 - Area 78.5 reservoirs

Start point
no restriction on spatial
average join: 14521.27
100%

nors cos > 0.9
pos dist < 0.5
average join: 14092.63
97.04%

nors cos > 0.9
pos dist < 0.1
average join: 13095.53
90.18%

nors cos > 0.9
pos dist < 0.04
average join: 5553.23
38.24%

Radius 2.5 - Area 19.625 reservoirs

Start point
no restriction on spatial
average join: 15011.42
100%

nors cos > 0.9
pos dist < 0.5
average join: 14743.55
98.21%

nors cos > 0.9
pos dist < 0.1
average join: 14698.51
97.91%

nors cos > 0.9
pos dist < 0.04
average join: 11608.62
77.33%

Ignoring GGX halfs cos and gradually concentrating area results in better merging percentage, better specular and better shadows but it gradually adds fireflies.